

Amjad Adi

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EDUCATION

Birzeit University

Bachelor's in Computer Engineering

West Bank, Palestine

Expected September 2027

- **GPA: 94.4 / 100**
- **Honors:** Consistently ranked among the top IT students in Birzeit University for 7 consecutive semesters.

EXPERIENCE

Competitive Programming & Technical Contests

West Bank, Palestine

Problem Solving

June 2024 – Present

- Solved **450+** problems on [Codeforces](#) and [LeetCode](#), applying **data structures, graph algorithms, dynamic programming, greedy algorithms, and computational complexity** to optimize time and space efficiency.
- Achieved **4th and 3rd Place** in two consecutive IEEE Head-to-Head competitions at Birzeit University, solving advanced logic problems under high-pressure constraints.
- Earned **4th Place** in the CES Club Team Competition, collaborating in a team of three to develop solutions for complex problems.

PROJECTS

Real-World Bookstore System | *Java, MySQL, JavaFX, AES, OOP*

January 2026

- Built a full-stack **bookstore management system** supporting **authentication and authorization, inventory management, order processing, reporting, and Admin/Customer workflows**, by designing a layered JavaFX application with MySQL integration and securing sensitive data using **AES-128 encryption**.

32-bit Five-Stage Pipelined RISC Processor | *Verilog, RTL, Computer Architecture*

July 2026

- Designed a **32-bit five-stage pipelined RISC processor** supporting **hazard detection, forwarding, stalling, and branch flushing**, by building the complete RTL datapath, control unit, and pipeline registers, verified through dedicated Verilog testbenches.

Low-Power Conditional-Capture Flip-Flop (CCFF) | *SPICE, VLSI Layout, 22nm CMOS*

January 2026

- Designed a **low-power conditional-capture flip-flop** in 22nm CMOS achieving a **42.5% reduction in dynamic power** versus a benchmark design, by implementing transistor-level SPICE circuits and a custom Electric VLSI layout with LVS/RC-parasitic and voltage/temperature sensitivity verification.

Single-Channel DMA Controller (UART → Memory) | *Verilog, FSM Design, Digital Design*

March 2026

- Designed a **DMA controller** that autonomously transfers data from a UART peripheral into memory without CPU intervention, by implementing a **3-state FSM** for byte-level read/write handshaking, verified end-to-end across independently-tested UART, Memory, and DMA modules.

Online Course Log Analyzer | *Bash, Shell Scripting, Linux*

November 2025

- Built a **menu-driven Bash log analyzer** providing 9 attendance-reporting modes (session counts, absentees, late/early arrivals, instructor and tool-usage statistics) for online course sessions, by chaining Linux text-processing commands (**grep, awk, cut, join, bc**) across two modular shell scripts with input validation.

Process Scheduling Simulator | *Java, JavaFX, Operating Systems, Data Structures*

January 2026

- Developed a **process scheduling simulator** supporting **Priority, Round Robin scheduling, I/O management, resource allocation, and deadlock recovery**, by implementing an event-driven Java engine with JavaFX visualization and performance analysis.

Sudoku Generator & Solver | *Python, CSP, Simulated Annealing, OOP*

December 2025

- Built a **9×9 and 16×16 Sudoku solver** comparing **CSP backtracking** against **Simulated Annealing**, by implementing MRV/MCV/LCV heuristics with forward checking and a cost-based SA optimizer to benchmark solving efficiency.

TECHNICAL SKILLS

Languages: Java, Python, C, SQL, Verilog HDL, Bash

Software Engineering: OOP, MVC, Multithreading, Socket Programming, Data Structures & Algorithms

Development: Android SDK, JavaFX, MySQL, SQLite, REST APIs, Git, Maven

Systems & Hardware: Computer Architecture, RTL/Digital Logic Design, FSM Design, SPICE, VLSI Layout, Linux